Foundation Trilogy

Selection, sharing and return

Complete Collected Volume

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Introduction

A structure that can be described need not be a structure that can be realized or shared. The Foundation papers carry this distinction through three stages: what a source and its noise let a reader select, what several readers actually share, and what a returned action changes while the old content and history are retained.

A shared algebra does not imply a shared state, and agreement does not establish correctness. A physically implemented return is stronger than updating an external estimate. The papers supply constructions and explicit conditions for these transitions, including preparation, calibration and receiver renewal rather than assuming them from an algebraic name.

One finite semiclassical mean-action scalar/JT construction joins old-content protection, the apparatus's own source and a second reading on the same retained history. An acquired record installs protection for old quantum content, changes that source, and receives its geometric response in a later X-basis reading of the old probe into a spent cell. Initial mass and geometric constraints are held fixed; task benefit and return occur in the same finite window. It does not simultaneously implement every Type III hull or a multi-reader public structure. Sustained thermal renewal is a further requirement, not part of this finite certificate.

Reading map

F-I *Foundation I: From Source to Manifestation. Accessible Algebras from a Source, Its Noise and a Readable Seed.* A source, its noise and a readable seed define an accessible hull. Selection and finite positive-energy preparation require their declared dynamical permissions.

F-II *Foundation II: Shared Manifestation. Shared Structure, Public Calibration and Different Histories.* Shared structure does not impose identical states or reliable common beliefs. Public references distinguish agreement, correctness and shared error.

F-III *Foundation III: The Perfuming Loop. Held Content, Learned Frames and Physical Return.* A return preserves old content while learned frames, physical action and source response change what can be read next. Receiver renewal is not renewal of an entire apparatus.

The papers retain their complete proofs, appendices and references, and their own pagination. The contents list PDF page numbers; PDF bookmarks also provide section-level navigation.